

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

SP6 - Financial Derivatives

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) Answer the following:

- i) A rolling call comes with barriers $H_1 > H_2 > \dots > H_n$ (all below the initial stock price) and strike prices $X_1 > X_2 > \dots > X_{n-1}$. This option starts as a European call with a strike price of X_0 . When the first barrier H_1 is hit, the strike price is rolled down to X_1 . In general, on hitting each barrier H_i , the strike is rolled down to X_i . The option knocks in when the last barrier H_n is hit. Replicate this option with a portfolio of down – and – out options.

[5]

- ii) Between a normal and a log normal model, which over prices out – of – the money calls and underprices out – of – the money puts on bonds?

[3]

- iii) Show that a cap is more valuable than an otherwise identical swaption.

[3]

- iv) Let $Y(t) = e^{X(t)}$, where $\{X(t), t \geq 0\}$ is a Brownian motion with drift m and variance s^2 . Show that

$$E\{Y(T)|Y(s)\} = Y(s)e^{(t-s)(\mu + \frac{\sigma^2}{2})}$$

[3]

- v) Suppose an asset follows Brownian motion and there are no interest rates. What can be said about the relative prices of out – of – the money American and European digital calls.

[3]

[17 marks]

Q.2) Prices for 6-month 60-strike European up-and-out call options on a stock S are available. Below is a table of option prices with respect to various H , the level of the barrier. Here, $S(0) = 50$.

H	Price of Up and Out Call
60	0
70	0.1294
80	0.7583
90	1.6616
μ	4.0861

Consider a special 6-month 60-strike European “knock-in, partial knock-out” call option that knocks in at $H_1 = 70$, and “partially” knocks out at $H_2 = 80$. The strike price of the option is 60. The following table summarizes the payoff at the exercise date:

H_1 Not Hit	H_1 Hit	
	H_2 Not Hit	H_2 Hit
0	$2 \times \max[S(0.5) - 60, 0]$	$\max[S(0.5) - 60, 0]$

Given this, calculate the price of the option.

[7 marks]

Q.3) The Reserve Bank of India has just issued, at par, a 5-year risk-free nominal callable bond that pays an annual coupon of 7.5%. The bond could be called in two years' time, at a price of ₹100. You are given the following information to calculate the value of the appropriate 2-year option on the bond:

- The annual effective 1-year and 2-year risk-free spot yields are 6% p.a. and 6.5% p.a. respectively.
- The forward yield is 8.3206% p.a.
- The forward yield volatility is 30%.
- The forward modified duration is 2.574.

i) Calculate the value of the callable bond. You may use the approximation: $\Phi(x) \approx 1/(1+e)$. [7]

ii) Explain the possible as to why Reserve Bank of India may have decided to issue the callable bond. [4]

[11 marks]

Q.4) A bank in the UAE has decided to hedge its position in a forward interest rate swap under which it will receive a fixed interest rate on a notional AED 1bn, starting in 2 years' time, with a further term of 3 years.

i) Explain why a short position in a forward bond might be considered an appropriate hedge for the bank's forward swap exposure. [2]

The bank identifies a 5-year UAE government bond with annual coupon rate of 10% and price AED 108 per AED 100 nominal. The bond just paid a coupon. The 1-year and 2-year effective risk-free spot rates are 6% p.a. and 7% p.a. respectively.

ii) Calculate the current forward price of the above bond in two years' time, per AED100 nominal.

The bank obtains the following volatility and correlation data:

- Standard deviation of price changes on the forward swap (fixed rate leg) = 5%
- Standard deviation of price changes on the forward bond = 6%
- Correlation coefficient between price changes on the above instruments = 0.9

[2]

iii) Calculate the nominal amount of the forward bond that would optimally hedge the bank's swap exposure. [2]

iv) State why this hedge is not a perfect hedge. [2]

v) The forward bond is arranged over the counter (OTC). Discuss the cashflows expected under the forward with and without a clearing house being used. [2]

[10 marks]

Q.5) A fund manager has a well – diversified portfolio that tracks the performance of a certain index and is worth EUR 275 million. The current value of the index is 1,100. The manager would like to buy portfolio insurance against a reduction of more than 5% in the value of the portfolio over the next year. The risk – free rate is 5% p.a.; the dividend yield on both the portfolio and the index is 3% while the market implied volatility for the index is currently 25% p.a.

- i) Calculate the cost of hedging the portfolio using European put options. [4]
 - ii) Describe alternative strategies involving European call options which would have the same effect as the options in (i). [4]
 - iii) Calculate the initial (delta) position if the manager sought to replicate the effect of the put options by investing part of the portfolio in risk-free securities. [3]
 - iv) Calculate the initial number of futures contracts required if, instead of risk-free securities in (iii), the manager decided to use 9-month index. [3]
- [14 marks]**

Q.6) A corporate bond issued by XYZ Ltd. has a face value of \$100 million, a maturity of 5 years, and pays an annual coupon of 6%. The current risk-free rate is 3% per annum. The probability of default for XYZ Ltd. is estimated at 4% per year, independent across years, with a recovery rate of 40% in case of default. A Credit Default Swap (CDS) is available on XYZ Ltd. with annual premium payments. Basis this,

- i) Explain the mechanics of a Credit Default Swap (CDS) and its role in transferring credit risk. [2]
 - ii) Derive the expected payoff to the protection buyer in case of default, given the recovery rate. [3]
 - iii) Calculate the fair annual CDS spread (premium rate) that equates the expected present value of premium payments to the expected present value of protection payments. Show all steps clearly. [7]
 - iv) Critically evaluate the factors that influence CDS pricing in practice, including market liquidity, counterparty risk, and correlation with broader credit markets. [3]
- [15 marks]**

Q.7) A fund manager purchases a payer swaption with option terms of 15 years and swap tenor of 5 years with a swap strike rate of 4%, a principal amount of ₹1000 Crores priced at ₹42Crores.

The zero – coupon bond prices for the interest rate for the next 20 years are specified as:

Base	Zero Coupon Bond Prices
1	0.99
2	0.97
3	0.96
4	0.94
5	0.92
6	0.9
7	0.87
8	0.85
9	0.82
10	0.79
11	0.76
12	0.74
13	0.7
14	0.68

15	0.65
16	0.62
17	0.59
18	0.56
19	0.53
20	0.49

- i) What does the term “moneyness of option” mean? Given the definition, is the above option “in the money”, “at the money” or “out of money”? [3]
- ii) What would be the implied volatility of the swaption with confidence interval of $\pm 2\%$. The market implied volatility of 15 – year swaption is given to be between 20% - 30%. [5]
- iii) The market yield curve has changed and the new zero – coupon bond prices are as given below. What is the revised price for this swaption? [6]

Base	Zero Coupon Bond Prices
1	1.00
2	1.00
3	0.99
4	0.98
5	0.96
6	0.94
7	0.91
8	0.88
9	0.84
10	0.81
11	0.76
12	0.72
13	0.68
14	0.64
15	0.60
16	0.55
17	0.51
18	0.46
19	0.43
20	0.38

- iv) The fund manager has priced the option after change in market curve to be ₹100 Crores. What could be the principal causes for difference (if any) between fund manager’s price and your price and how can you allow for this difference in your model. [2]
[16 marks]

Q.8) You are an investment actuary working in an insurance company. The product team has referred a parametric insurance policy to you as they felt the product construct is better understood by an actuary who is experienced on the investment side. The product is a livestock-mortality insurance product for small farmers. The policy pays a predetermined

amount whenever a satellite-derived vegetation-index threshold is breached over a 6-month period, without requiring individual farm loss verification.

- i) Explain how this parametric insurance contract can be viewed as a path-dependent derivative written on a weather/environmental index. What is the payoff of a binary parametric policy? [4]
- ii) Describe how you would approach pricing this parametric product, drawing on derivative pricing concepts such as calibration of index dynamics to historical data, the treatment of jump risk and structural breaks like climate change trends. [3]
- iii) Explain how you could use standard derivatives to hedge the exposure to this product. [3]

[10 marks]
